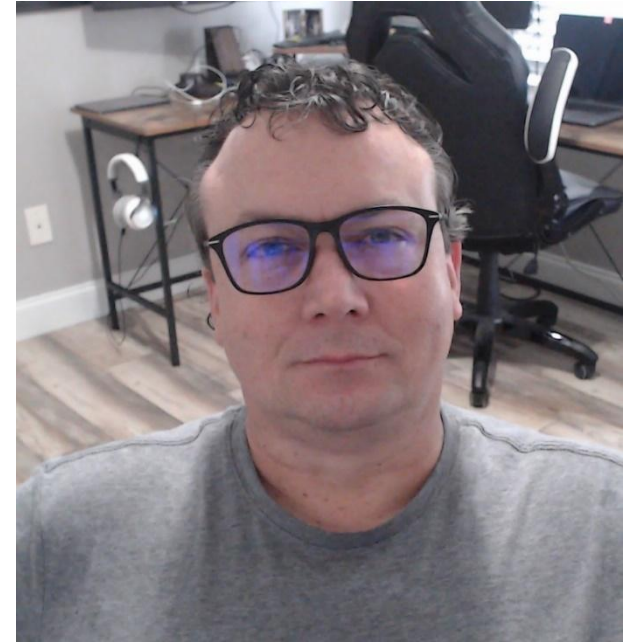




SQL Server

Foundations and SSMS



Instructor: Brian Dietrick



SQL Server Series

1. Foundations and SSMS
2. Installation, Modeling, and Development
3. Core Administration and Operations
4. Performance Tuning and Deep Dive



1. Foundations and SSMS

- Day 1 – Platform and Instance Foundations
- Day 2 – Storage, Memory, and Transaction Log Internals
- Day 3 – SSMS and Observability Tooling
- Day 4 – Security and Support T-SQL Essentials



Day 1 Agenda

- SQL Server Support
- SQL Server Feature Releases – selected features
- Platform overview
- Database Engine and SQL Server Agent
- SSIS, SSRS, SSAS overview
- Service startup dependencies and account context
- Instance concepts
- Default vs. named instances
- Ports, protocols, SQL Browser, and connectivity
- Lab and knowledge check



Microsoft SQL Server

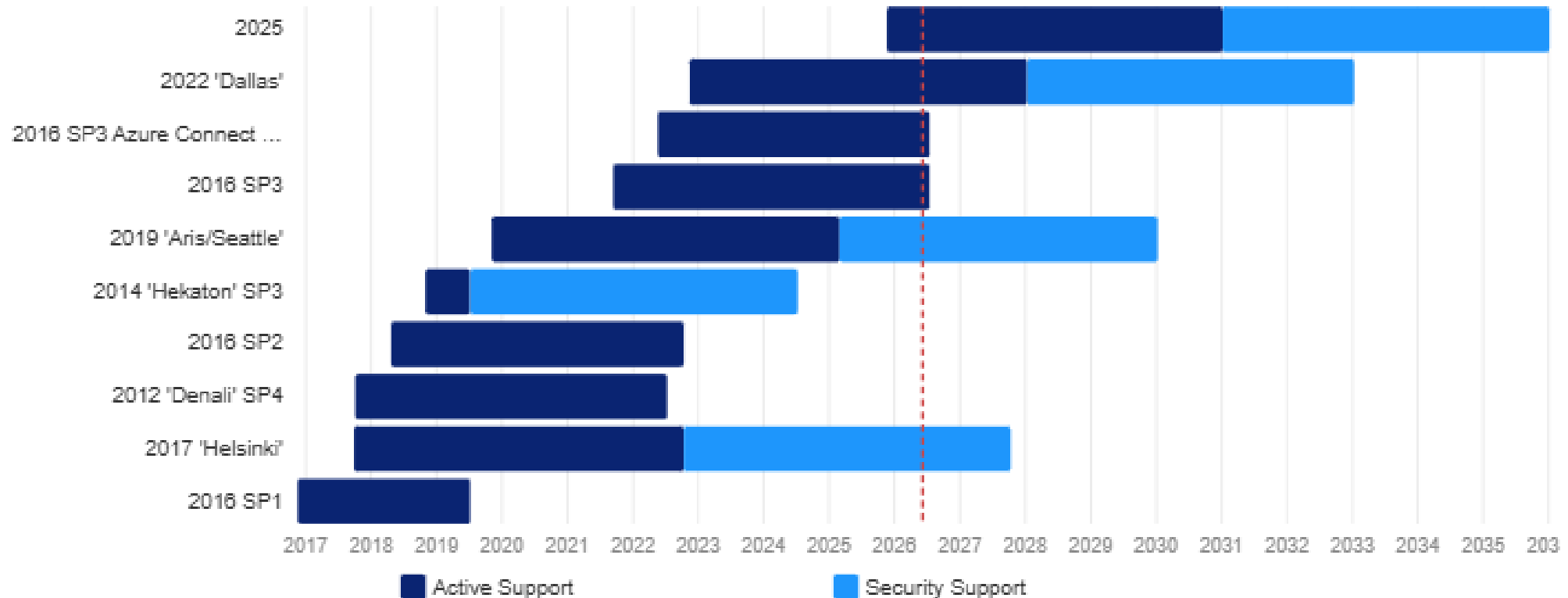
DATABASE

MICROSOFT

 Last updated on 03 June 2026



[SQLServer](#): Microsoft SQL Server is a relational database management system developed by Microsoft.



SQL Server 2016: *The foundation for the modern engine*

Key Features Added

- **Query Store**

- Tracks query text, plans, and performance metrics for regression troubleshooting.

- **Built-in JSON support**

- Parse, import, export JSON; FOR JSON, OPENJSON

- **Security stack overhaul:**

- Always Encrypted
- Dynamic Data Masking
- Row-Level Security

- **TempDB improvements**

- Multiple files configured automatically during setup.



SQL Server 2017: *Cross-platform + adaptive performance*

Key features Added

- **Linux**
 - Run SQL Server on Linux
- **Docker support**
 - Run SQL Server in a container
- **Graph database**
 - (nodes, edges, MATCH)
- **Python Machine Learning Services**
 - Execute python inside SQL Server
 - Eliminate data movement
 - Operationalizes ML
- **Memory Grant Feedback**
 - Addresses over and under allocation of memory
- **Automatic Plan Correction**
 - Self healing performance
- **Resumable online index rebuilds**
 - Survives interruptions
 - Fits into short maintenance windows
 - Reduces log pressure
 - Avoids wasted work



SQL Server 2019: *Big Data + Intelligent Query Processing*

Key Features Added

- **Persisted Version Store**
 - Eliminates read blocks waiting on writes
 - Eliminates write blocks waiting on reads
- **Accelerated Database Recovery (ADR)**
 - near-instant rollback
- **UTF-8 support**
 - Improved storage efficiency over the previous UTF-16
 - Improves performance for text-heavy workloads
- **Virtual Data**
 - Access external data sources from T-SQL (Oracle, Teradata, MongoDB, Hadoop, Cosmos, Flat files, etc.)
- **Scalar UDF inlining**
 - Massive performance improvements for UDF-heavy workloads
- **Batch mode on rowstore**
 - Lowers CPU utilization for SQL Servers default traditional storage format



SQL Server 2022: *Cloud integration + governance + IQP*

Key Features Added

- **Azure Synapse Link**
 - Eliminate ETL for analytics and near zero impact on OLTP
- **Microsoft Purview integration**
 - Centralizes governance, security, and compliance
- **Parameter Sensitive Plan (PSP) optimization**
 - Huge performance improvement for stored procedures
- **Ledger tables**
 - Blockchain-style tamper evidence
 - Detects unauthorized changes even by privileged user
- **Contained Availability Groups**
 - Automatic migration of server level objects when you migrate a user database
- **TempDB concurrency improvements**
 - Dramatic performance improvement for tempDB heavy workloads



SQL Server 2025: *AI-native engine + JSON overhaul*

- **Key Features Added**

- **Native JSON data type**
 - not NVARCHAR
- **JSON indexes + validation**
- **Full Regular Expression support**
- **BASE64 encode/decode functions**
- **PRODUCT aggregate**
- **External REST invocation**
(sys.sp_invoke_external_rest_endpoint)
- **Vector data type + vector indexes (DiskANN)**
- **AI integration with Azure AI Foundry, Azure OpenAI, Semantic Kernel, LangChain, Ollama**
- **Improved ADR, locking, and availability**



SQL Server as a Data Platform

- Relational database engine
- Scheduling and automation
- Data integration
- Reporting
- Analytics and semantic modeling
- Administrative and development tooling

Major SQL Server Components

- Database Engine
- SQL Server Agent
- SQL Server Integration Services (SSIS)
- SQL Server Reporting Services (SSRS)
- SQL Server Analysis Services (SSAS)

Services vs. Features

- A service runs as a Windows service
- A feature may be a tool, library, runtime, or optional capability
- Not every installed SQL Server component is always running
- Troubleshooting starts with knowing what should be running

SQL Server 2016 Improvements

- Query Store
- Always Encrypted
- Temporal tables
- R Services
- PolyBase

Typical Deployment Patterns

- Standalone SQL Server
- Failover Cluster Instance (FCI)
- AlwaysOn Availability Groups

SQL Server Tools Ecosystem

- SQL Server Management Studio (SSMS)
- SQLCMD
- PowerShell modules
- SQL Server Data Tools (SSDT)

```
on PirateDevBox C:\code\demo  
+ > sqlcmd -S localhost\sql2016 -E -Q "SELECT @@VERSION;"
```

```
-----  
Microsoft SQL Server 2016 (SP3-GDR) (KB5089271) - 13.0.6490.1 (X64)  
Apr  8 2026 08:27:03  
Copyright (c) Microsoft Corporation  
Developer Edition (64-bit) on Windows 10 Pro 10.0 <X64> (Build 26200: ) (Hypervisor)
```



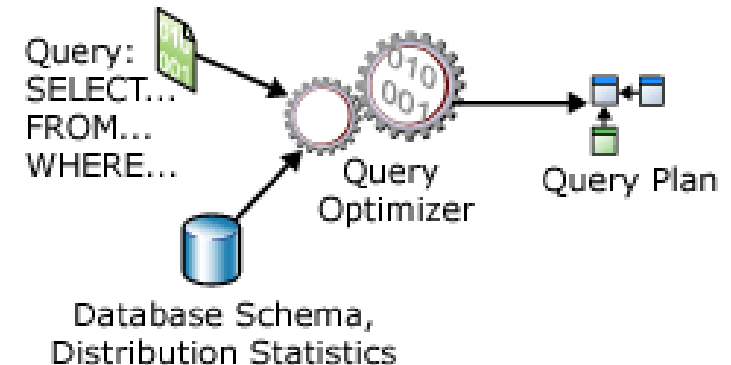
Database Engine Overview

- Core relational engine
- Storage engine + query processor (relational engine)
- Owns transactions, locking, logging, recovery, and execution



Storage Engine and Query Processor

- Storage engine:
 - Reads data pages from disk, performs index seeks and table scans
 - Writes updates to data pages
 - Coordinates physical retrieving and navigating of data
- Query processor:
 - Parses query
 - Optimizes query based on cardinal estimates
 - Builds an execution plan
 - Executes the plan
- Both are needed for query execution



What the Database Engine Manages

- Transactions
- Locking and concurrency
- Recovery
- Buffer pool
- Plan cache
- tempdb
- I/O scheduling

SQL Server Agent Overview

- Job scheduling and automation engine
- Runs scheduled or event-driven tasks
- Supports alerts and operators
- Depends on the Database Engine service

What SQL Server Agent Can Run

On-slide content

- T-SQL job steps
- SSIS package execution
- PowerShell steps
- CmdExec steps
- Alerts and notifications

...proxy accounts are important.



SSIS

- Data movement and transformation platform
- Often used for ETL workflows
- Common in batch and integration scenarios

SSRS

- Reporting platform
- Delivers operational and formatted reports
- Often tied to subscriptions, schedules, and business reporting
- Export data in several formats – Excel, HTML, PDF, etc.
- SSRS is not the Database Engine itself.
- Report issues can involve data, permissions, rendering, or the report service stack.
- SSRS is not the Database Engine itself.



SSAS

- Analytical and semantic modeling platform
 - Supports tabular and multidimensional models
 - Used for business intelligence and analytics scenarios
-
- Largely replaced by PowerBI

Service Account Context

- Service accounts define execution identity
- Identity affects file access
- Identity affects network access
- Identity affects delegation and double-hop scenarios
- Identity affects what jobs and services can do

SQL Server Service Permissions

- Local file system access
- Network share access
- Service startup permissions
- Kerberos and delegation implications

A service account with insufficient access can cause failures even when T-SQL is valid.



SQL Server Agent Account and Proxies

- Agent may run under a different service account
- Some job steps need proxy credentials
- Execution context matters for PowerShell, CmdExec, and external access

"...the script works manually but fails as a job."

Startup Dependencies

- Database Engine service must be online for many dependent operations
- Agent depends on the engine
- Supporting OS services may also matter
- Startup order affects availability and troubleshooting

Why Configuration Manager Matters

- Use SQL Server Configuration Manager for service identity changes
- Use it for protocols and network settings
- Use it for startup parameters
- Avoid generic service tools when SQL-aware settings matter



Best Practices for Services and Accounts

- Use least privilege
- Use dedicated service accounts where appropriate
- Document account purpose and scope
- Avoid unnecessary local administrator rights
- Validate network and file access before production rollout

SQL Server Instance Concepts

- Multiple instances can exist on one server
- Each instance is a separate SQL Server installation context
- Each instance has its own system databases and settings
- Each instance is an administrative boundary

What Each Instance Possesses

- Its own service name
- Its own system databases
- Its own logins and instance-level configuration
- Its own memory and CPU settings
- Its own network identity and port configuration



Instance-Level vs. Database-Level Settings

- Instance level:
 - memory configuration
 - server collation
 - logins
 - server-wide settings
- Database level:
 - database collation
 - recovery model
 - file layout
 - users and objects



Instance Isolation

- Performance boundary
- Security boundary
- Administrative boundary
- Operational flexibility

...tradeoffs: isolation helps, but instances still compete for the host's physical resources



Default vs. Named Instances

- Default instance
- Named instance
- Naming affects connection syntax
- Naming often affects port discovery behavior

...Default instance usually connects with just the server name.

...Named instance connects with `ServerName\InstanceName` unless using an explicit port



Default Instance Basics

- Usually associated with TCP 1433
- Uses the default service name pattern
- Common for simpler deployments

...Explain that TCP 1433 is the usual default, not a law of nature.

...If changed by administrators, clients must match the actual configuration.



Named Instance Basics

- Often uses dynamic TCP ports by default
- Can be configured with a fixed port
- Often relies on instance-to-port discovery

Protocols Managed by SQL Server

- Shared Memory
- TCP/IP
- Named Pipes
- Managed through SQL Server Configuration Manager

SQL Browser

- Helps resolve named instances to ports
- Listens on UDP 1434
- Important for many named instance connection patterns

When SQL Browser Is Required

- Often needed for named instance discovery
- Less important when clients already know the fixed port
- Not typically needed for a default instance on a known port

`ServerName,Port` can bypass name-to-port discovery in many



Client Connectivity Basics

- Server name
- Instance name or port
- Authentication mode
- Network reachability
- Protocol compatibility

If a client cannot connect, start with identity, address, endpoint, and protocol.



Connection String Examples

- Default instance: `SQLLAB01`
- Named instance: `SQLLAB01\HR`
- Explicit port: `SQLLAB01,51433`
- Authentication details depend on the client and provider

Authentication Modes

- Windows Authentication
- SQL Authentication
- The chosen mode changes login behavior and troubleshooting

Common Connectivity Issues

- Wrong server or instance name
- Wrong or blocked port
- SQL Browser unavailable
- TCP/IP disabled
- Authentication mismatch
- Service stopped
- Firewall restrictions

Troubleshooting Flow

- Is the service running?
- Is the target server or instance name correct?
- Is the expected protocol enabled?
- Is the expected port reachable?
- Is SQL Browser needed?
- Is authentication correct?

Demo Setup Overview

- Connect in SSMS
- Identify the instance
- Review system databases
- Create the demo database
- Query support-oriented demo data

...demo flow and scripts.



Demo Script 1: Identify the Instance

```
SELECT
  @@SERVERNAME AS server_name,
  @@SERVICENAME AS service_name,
  @@VERSION AS version_string;
```

...confirm where you are connected.

`MSSQLSERVER` typically indicates the default instance service.

What would expect to see for a named instance?



Demo Script 2: Review Databases

```
SELECT
    name,
    database_id,
    state_desc,
    recovery_model_desc
FROM sys.databases
ORDER BY name;
```

Speaker notes

- Show the system databases and explain their names without diving deeply yet.
- Production-safety note: This metadata query is lightweight and generally safe in production, but it should still be run with read-only intent and interpreted carefully during incidents.

Demo Script 3: Inspect Instance Properties

SELECT

```
SERVERPROPERTY('MachineName') AS machine_name,  
SERVERPROPERTY('ServerName') AS server_name,  
SERVERPROPERTY('InstanceName') AS instance_name,  
SERVERPROPERTY('Edition') AS edition,  
SERVERPROPERTY('ProductVersion') AS product_version,  
SERVERPROPERTY('ProductLevel') AS product_level;
```

Demo Script 4: Create the Demo Database

```
IF DB_ID('CourseDemo_Module1') IS NOT NULL
BEGIN
    ALTER DATABASE CourseDemo_Module1 SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE CourseDemo_Module1;
END;
GO

CREATE DATABASE CourseDemo_Module1;
GO

USE CourseDemo_Module1;
GO

CREATE TABLE dbo.SupportTickets
(
    TicketId int IDENTITY(1,1) PRIMARY KEY,
    SystemName nvarchar(50) NOT NULL,
    IssueType nvarchar(50) NOT NULL,
    PriorityLevel varchar(10) NOT NULL,
    OpenedBy nvarchar(50) NOT NULL,
    OpenedDate datetime2 NOT NULL DEFAULT SYSUTCDATETIME()
);
```

Demo Script 5: Populate and Query the Dataset

```
INSERT INTO dbo.SupportTickets (SystemName, IssueType, PriorityLevel, OpenedBy)
VALUES
('SQLLAB01', 'Login Failure', 'High', 'Alex'),
('SQLLAB01\\HR', 'Named Instance Not Found', 'Medium', 'Jordan'),
('SQLLAB02', 'Service Will Not Start', 'High', 'Sam'),
('SQLLAB03', 'Port Blocked', 'Medium', 'Taylor'),
('SQLLAB04', 'Agent Job Failed', 'Low', 'Morgan');
GO
```

```
SELECT TOP (5)
    TicketId,
    SystemName,
    IssueType,
    PriorityLevel,
    OpenedBy,
    OpenedDate
FROM dbo.SupportTickets
ORDER BY TicketId
```

Demo Script 6: Database-Level Example

```
SELECT
    name,
    compatibility_level,
    collation_name
FROM sys.databases
WHERE name = 'CourseDemo_Module1';
```

Student Handout: Key Terms

- SQL Server platform
- Database Engine
- SQL Server Agent
- Service account
- Instance
- Default instance
- Named instance
- SQL Browser
- TCP/IP
- Windows Authentication



Student Handout: Quick Reference Table

Topic	Quick reference
Default instance	Commonly connects as <code>ServerName</code>
Named instance	Commonly connects as <code>ServerName\\InstanceName</code>
Default TCP port	Usually 1433
SQL Browser	Helps map named instance to port
Protocols	Shared Memory, TCP/IP, Named Pipes
First connectivity checks	Service, name, port, protocol, auth

Lab Introduction

- Goal: identify instance behavior and reason through connectivity issues
- Work in SSMS
- Use provided scripts
- Record observations, not just outputs

...the lab is about interpretation, not speed.

...write down why a connection pattern would succeed or fail.



Lab Tasks

- Connect to the assigned instance
- Run the environment verification script
- Determine default vs. named instance behavior
- Create the demo database
- Review `sys.databases`
- Evaluate support scenarios
- Document instance-level vs. database-level facts

Lab Script: Environment Verification

```
SELECT
  @@SERVERNAME AS server_name,
  @@SERVICENAME AS service_name,
  SERVERPROPERTY('Edition') AS edition,
  SERVERPROPERTY('ProductVersion') AS product_version;
```

...what the service name suggests about instance type.



Lab Script: Scenario Table Setup

```
IF DB_ID('CourseDemo_Module1') IS NOT NULL
BEGIN
    ALTER DATABASE CourseDemo_Module1 SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE CourseDemo_Module1;
END;
GO

CREATE DATABASE CourseDemo_Module1;
GO

USE CourseDemo_Module1;
GO

CREATE TABLE dbo.InstanceScenarios
(
    ScenarioId int IDENTITY(1,1) PRIMARY KEY,
    ScenarioName nvarchar(100) NOT NULL,
    ConnectionExample nvarchar(100) NOT NULL,
    LikelyCause nvarchar(200) NOT NULL
);
GO

INSERT INTO dbo.InstanceScenarios (ScenarioName, ConnectionExample, LikelyCause)
VALUES
('Default instance connection', 'SQLLAB01', 'Engine reachable on expected endpoint'),
('Named instance connection', 'SQLLAB01\HR', 'Client may need SQL Browser or a fixed port'),
('Explicit port connection', 'SQLLAB01,51433', 'Used when named instance discovery is bypassed'),
('Authentication issue', 'SQLLAB02', 'Login mode or permissions mismatch'),
```

Lab Script: Review Scenarios

```
USE CourseDemo_Module1;  
GO  
  
SELECT TOP (10)  
    ScenarioId,  
    ScenarioName,  
    ConnectionExample,  
    LikelyCause  
FROM dbo.InstanceScenarios  
ORDER BY ScenarioId;
```

...which rows might require SQL Browser, which ones may involve authentication, and which ones suggest a protocol or firewall problem.



Lab Validation Checklist

- Can you identify default vs. named instance behavior?
- Can you explain what SQL Browser does?
- Can you list server, instance or port, and authentication as connection elements?
- Can you explain why Agent depends on the Database Engine?
- Can you separate instance-level and database-level settings?



Knowledge Check

- What is the Database Engine responsible for?
- Why is SQL Server a platform?
- What is the difference between a service and a feature?
- What is an instance?
- When does SQL Browser matter?

Knowledge Check Answer Highlights

- Database Engine: transactions, locking, logging, recovery, execution
- SQL Server is a platform because it includes multiple services, tools, and feature areas
- Service vs. feature: runtime process vs. installed capability
- Instance: isolated SQL Server runtime and configuration context
- SQL Browser: assists named instance port discovery



Wrap-Up

- SQL Server is more than a database process
- Components have distinct responsibilities
- Service identity and startup order matter
- Instance concepts shape administration and connectivity
- Ports, protocols, and SQL Browser drive many connection outcomes



Appendix A - Student Handout Notes

Summary bullets

- SQL Server 2016 includes a relational engine plus supporting platform services and tools.
- Database Engine and SQL Server Agent are the most operationally important components for many support roles.
- SSIS, SSRS, and SSAS should be recognized even if not administered in depth.
- Service accounts affect what SQL Server can access and how jobs run.
- A single server can host multiple SQL Server instances.
- Default and named instances differ in naming and often in port discovery behavior.
- SQL Browser helps named instance resolution.
- Connectivity troubleshooting should start with service state, target name, protocol, port and authentication



Appendix B - Detailed Lab Handout

Lab goal

- Identify a SQL Server instance, distinguish default and named instance behavior, and reason through basic connectivity and service-account-related issues.

Prerequisites

- Access to SSMS connected to a SQL Server 2016 lab instance.
- Permission to create a training database in the lab environment.
- Instructor-provided lab server names and example connection patterns.
- SQL Server Agent installed in the lab where possible.

Tasks

1. Connect to the assigned SQL Server instance in SSMS.
2. Run the environment verification script and record server name service name